

★ SERIES + PARALLEL RESISTORS ★

GS | Ch03 Circuits + Chem A | Topic 1 of 12

► Meaning in one line

Add vs reciprocal , same I vs same V.

► Key points

Point	Example
Series: $R_s = R_1 + R_2$ + same current.	lights in a row.
Parallel: same voltage + $\frac{1}{R_p} = \text{sum}$.	home wiring.

★ IMPORTANT FOR EXAM

Power + energy + safety

⚠ TRAP - read twice

Add-all + 'resistors asked' + mode-miss + same-miss

✓ **Try yourself:** Mode + same + sum + combo? **Answer:** Series vs parallel.

★ ELECTRIC POWER + ENERGY UNIT ★

GS | Ch03 Circuits + Chem A | Topic 2 of 12

► Meaning in one line

$P = VI$, kWh commercial.

► Key points

Point	Example
Power $P = V \times I = I^2 R$ + watt.	100 W bulb.
Energy kWh = units + bill = units x rate.	2 kW x 3 h = 6 units.

★ IMPORTANT FOR EXAM

Conductors + magnets + safety

⚠ TRAP - read twice

Watt-only + 'power asked' + formula-miss + unit-miss

✓ **Try yourself:** VI + kWh + rate + bill + math? **Answer:** P + units.

★ CONDUCTORS + MAGNETS + ELECTROMAGNET ★

GS | Ch03 Circuits + Chem A | Topic 3 of 12

► Meaning in one line

Flow vs block , poles , coil-current.

► Key points

Point	Example
Conductors flow + insulators block.	copper vs rubber.
Magnet N/S + like repels + electromagnet = coil + current + soft core.	crane + bell.

★ IMPORTANT FOR EXAM

Safety + matter

⚠ TRAP - read twice

Iron-only + 'magnet asked' + pole-miss + current-miss

✓ **Try yourself:** Flow + poles + current + field + sort? **Answer:** Conductor + magnet.

★ DOMESTIC SAFETY + STATES OF MATTER ★

GS | Ch03 Circuits + Chem A | Topic 4 of 12

► Meaning in one line

Fuse/MCB/earth , solid/liquid/gas.

► Key points

Point	Example
Safety: fuse melts + MCB trips + earth guards.	3-pin plug.
States: solid fixed + liquid flows + gas fills.	ice/water/steam.

★ IMPORTANT FOR EXAM

Changes + physical/chemical

⚠ TRAP - read twice

Wire-only + 'safety asked' + device-miss + state-miss

✓ **Try yourself:** Protect + shape + device + state? **Answer:** Fuse + states.

★ CHANGES OF STATE + PHYSICAL/CHEMICAL ★

GS | Ch03 Circuits + Chem A | Topic 5 of 12

► Meaning in one line

Melt/freeze/boil/condense , reversible vs new-substance.

► Key points

Point	Example
States change: melt/freeze/boil/condense/sublime.	camphor sublimates.
Physical reversible + chemical new substance.	melt vs burn/rust.

★ IMPORTANT FOR EXAM

Atom + element/compound

⚠ TRAP - read twice

Change-only + 'change asked' + reverse-miss + new-miss

✓ **Try yourself:** Reverse + new + heat + change + sort? **Answer:** Physical vs chemical.

★ ATOM + ELEMENT/COMPOUND/MIXTURE ★

GS | Ch03 Circuits + Chem A | Topic 6 of 12

► Meaning in one line

Particle , pure vs combined vs mixed.

► Key points

Point	Example
Atom smallest + molecule bonded + ion charged.	Na+ / Cl-.
Element pure-one + compound fixed-ratio + mixture separable.	air mixture.

★ IMPORTANT FOR EXAM

Mixtures + subatomic

⚠ TRAP - read twice

Mix-only + 'matter asked' + ratio-miss + separable-miss

✓ **Try yourself:** Particle + ratio + separable + matter + sort? **Answer:** Atom + element.

★ MIXTURE TYPES + SUBATOMIC TRIO ★

GS | Ch03 Circuits + Chem A | Topic 7 of 12

► Meaning in one line

Uniform vs non-uniform , p/n/e.

► Key points

Point	Example
Homogeneous uniform + hetero non-uniform.	salt solution vs sand.
Proton + / neutron 0 / electron - + mass.	e lightest.

★ IMPORTANT FOR EXAM

Atomic numbers + isotopes

⚠ TRAP - read twice

Mix-only + 'tiny asked' + uniform-miss + charge-miss

✓ **Try yourself:** Uniform + charge + mass + tiny + sort? **Answer:** Mixture + p/n/e.

★ ATOMIC NUMBER + ISOTOPES ★

GS | Ch03 Circuits + Chem A | Topic 8 of 12

► Meaning in one line

Z = protons , A = p , n , same- Z .

► Key points

Point	Example
Z = protons = electrons + A = protons + neutrons.	carbon Z 6.
Isotopes: same Z + diff neutrons.	C-12/C-14 + U-235 fuel.

★ IMPORTANT FOR EXAM

Periodic + acids

⚠ TRAP - read twice

Mass-only + 'atom asked' + Z -miss + same-miss

✓ Try yourself: Z + A + same + identity? **Answer:** Z/A + isotopes.

★ PERIODIC TABLE + ACID/BASE ★

GS | Ch03 Circuits + Chem A | Topic 9 of 12

► Meaning in one line

Periods/groups , sour/soapy.

► Key points

Point	Example
Periods rows + groups columns + metals left.	group 1 alkali.
Acid sour H^+ + base bitter OH^- + soapy.	lemon vs soap.

★ IMPORTANT FOR EXAM

pH + indicators

⚠ TRAP - read twice

Name-only + 'table asked' + trend-miss + ion-miss

✓ **Try yourself:** Rows + ions + sour + table + sort? **Answer:** Periodic + acid/base.

★ pH SCALE + INDICATORS ★

GS | Ch03 Circuits + Chem A | Topic 10 of 12

► Meaning in one line

0-14 , 7 neutral , colour flips.

► Key points

Point	Example
pH below 7 acid + 7 neutral + above 7 base.	blood 7.4.
Litmus red/blue + phenol pink in base + methyl orange.	lab tests.

★ IMPORTANT FOR EXAM

Neutralization + common acids

⚠ TRAP - read twice

Colour-only + 'pH asked' + number-miss + flip-miss

✓ **Try yourself:** Number + flip + scale + colour + read? **Answer:** pH + indicator.

★ NEUTRALIZATION + COMMON ACIDS ★

GS | Ch03 Circuits + Chem A | Topic 11 of 12

► Meaning in one line

Acid , base = salt , water , names.

► Key points

Point	Example
Neutralization: acid + base = salt + water.	antacid relief.
Common acids: HCl/H ₂ SO ₄ /HNO ₃ + bases NaOH/CaO.	battery + cleaner.

★ IMPORTANT FOR EXAM

Salts + daily-pH

⚠ TRAP - read twice

Mix-only + 'acid asked' + product-miss + name-miss

✓ **Try yourself:** Product + name + use + acid? **Answer:** Salt-water + names.

★ COMMON SALTS ★

GS | Ch03 Circuits + Chem A | Topic 12 of 12

► Meaning in one line

NaCl/washing/baking/plaster , uses.

► Key points

Point	Example
Table NaCl + washing Na ₂ CO ₃ + baking NaHCO ₃ .	kitchen + clean.
Plaster POP + gypsum.	casts + walls.

★ IMPORTANT FOR EXAM

Till end: daily-pH + metals!

⚠ TRAP - read twice

White-only + 'salt asked' + formula-miss + use-miss

✓ **Try yourself:** Formula + use + salt + daily? **Answer:** Salts + uses.